Preventing disease emergence following eradication: application to mpox

Abstract

The success of the vaccination campaign against smallpox led to its eradication and the discontinuation of vaccination. The subsequent decline in the level of population immunity against related poxviruses such as mpox has led to an increase both in the number of zoonotic spillovers of these viruses as well as an increase in the extent of their transmission between humans, raising the specter of the emergence of mpox as a global threat. We use mathematical models to explore how the probability of mpox emergence increases as the level of herd-immunity to smallpox wanes. We find that: (i) changes in the transmissibility have a much larger contribution compared with changes in spillover rates for the probability of evolutionary emergence; (ii) the vaccination threshold need to prevent evolutionary emergence is potentially much lower than that needed for eradication after evolutionary emergence and adaptation. Our results highlight the importance of developing and implementing an effective strategy to prevent the evolutionary emergence of mpox.

I understand population immunity is important to prevent sustained transmission, but why would number of zoonotic spillovers depend on population immunity? Shouldn't it be more dependent on contact rates between reservoir host and humans?

After reading the whole paper I understood this point, but it might be helpful for the reader if you can mention "successful" zoonotic spillovers that actualy causes an infection.

1 Introduction

Our understanding of the use of vaccines and other intervention measures for the control and eradication of highly transmissible infectious pathogens was facilitated by the insights provide by relatively simple mathematical models of disease transmission and control [1, 15]. These models suggest that the control of a pathogen depends on its basic reproductive number R_0 which equals the number of secondary infections arising from an infected individual introduced into a completely susceptible population [12]. Disease eradication requires vaccine coverage to exceed a fraction $1 - \frac{1}{R_0}$ of the population. The successful implementation of vaccination strategies inspired by consideration of the above heuristic led to the global eradication of two diseases: smallpox in humans (1980) [2] and rinderpest in cattle (2011) [16]. Because smallpox lacks a zoonotic reservoir (humans are the sole hosts for virus), vaccination against smallpox was discontinued following eradication [4, 7, 19].

reference numbers!

While there is no zoonotic reservoir in which smallpox circulates, related orthopoxviruses, which we generically refer to as mpox, do circulate in a number

of mammalian hosts and have the ability to infect humans [4, 10]. Historically, mpox did not pose a threat to human populations as there were relatively few zoonotic cases and very limited human-human spread from these spillovers [5]. This was due to widespread immunity to smallpox which conferred cross-reactive protection against mpox [10]. The cessation of mass vaccination against smallpox and the consequent decline in the level of population immunity has potentially opened a niche for the emergence of mpox [3, 14, 18?]. reference!

In this paper we develop a simple framework analogous to that used for disease eradication to explore how waning herd immunity can facilitate the adaptation of mpox to humans and its emergence as a highly infectious disease. We explore how the probability of evolutionary emergence of mpox increases over time, and how interventions can prevent or at least substantially delay the process of evolutionary emergence. Our goal is to develop an intuitive understanding of the key factors that govern the evolutionary emergence of zoonoses following eradication of human pathogens.

Results and Discussion

We clarify our use of R_0, the basic reproductive number and R_e, the effective reproductive number. R_0 equals the typical number of secondary cases arising from introduction of an infected individual into a wholly susceptible population. The effective reproductive number R_e takes into account the extent of immunity in the population.

Outline of the process of emergence

In Fig 1 we outline the stages for the emergence of a pathogen such as mpox along the lines described in [17]. The first stage is cross-species transmission, termed spillover, that results in the introduction of infections into the human population. These spillover events can lead to chains of transmission of the pathogen in the population. If the effective reproductive number $R_e < 1$, which is the scenario we focus on in this paper, then these chains of transmission will stutter to extinction. Evolution of the pathogen during these stuttering chains of transmission can result an increase in both R_0 and R_e and, if the resulting $R_e > 1$, this would allow mpox to persist in the human population. If R_e has increased to above 1 prior to mpox evolving then further evolution is not needed for it to persist in the human population [23]. The final stage is the further adaptation of mpox for transmission in the human population. We expect this stage to end when the pathogen has reached a fitness peak for transmission in the human population, after which further adaptive evolution will stop or be greatly reduced – for mpox, this might occur when the evolved mpox variant has a basic reproductive number close to that of smallpox.

maybe write someone et al. [ref #]

Do not completely understand the logic of this sentence, Are you saying that mpox can't have a higher R0 than smallpox? Why?

We consider how the progress through these stages is affected by the waning of population immunity.

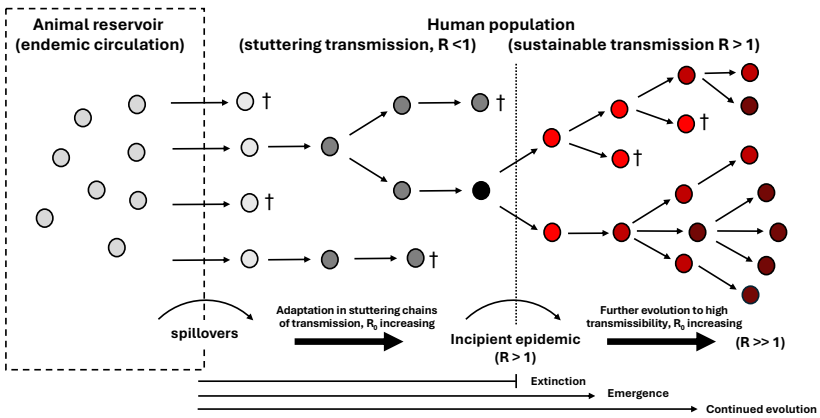

Figure 1: Schematic showing the stages for the emergence of a pathogen. Initially the pathogen causes infections the zoonotic reservoir (circles) and these occasionally spill over into the human population. If the initial reproductive number, R_e, of the pathogen in humans is less than one (grey circles) it can nevertheless cause chains of transmission that stutter to extinction. During these chains of transmission R can increase (depicted by darker color) and if R becomes greater than one (red circles) it can persist in the human population. Adaptive evolution can increase R further ending when it has reached a fitness peak in humans (darker red circles).

light grey

dark

How R is different from Re and R0? The first result section may need to define what is R?

Waning immunity and the reproductive number of mpox

We now consider the waning of population immunity to smallpox (and mpox) and the concomitant increase in the R_e of mpox. If immunity to smallpox is lifelong and all individuals were immune to smallpox when vaccination ended, then we expect population level immunity to smallpox $F(t)$ to decline with the fraction of the population born prior to the time when smallpox vaccination ceased (circa 1980):

after?

$$F(t) = \left(1 - \int_{a=0}^{t} f(a)da\right) \tag{1}$$

where t is the time at which vaccination was stopped and $f(a)$ is the fraction of population of age a. In Fig 2 we use estimates for the demographics of the Democratic Republic of Congo (DRC) to get an idea of how the frequency of individuals with smallpox immunity declines and the reproductive number of mpox increases over time [20]. The reader is referred to [**?**] for a more nuanced picture of the waning of immunity worldwide.

The effective reproductive number of mpox, R_e is related to $F(t)$ by:

$$R_e(t) = R_0(1 - cF(t)) \tag{2}$$

where c is the extent to which smallpox immunity protects against mpox. It is clear that the R_e of mpox was less than one just after the eradication of smallpox around 1980. Early contact tracing studies suggested R_0 of mpox was

Just curious, does 'c' need to be constant with time? in case of adaptive evolution, as addressed in the next section, if antigenic shift happens, the cross-immunity can change, right?

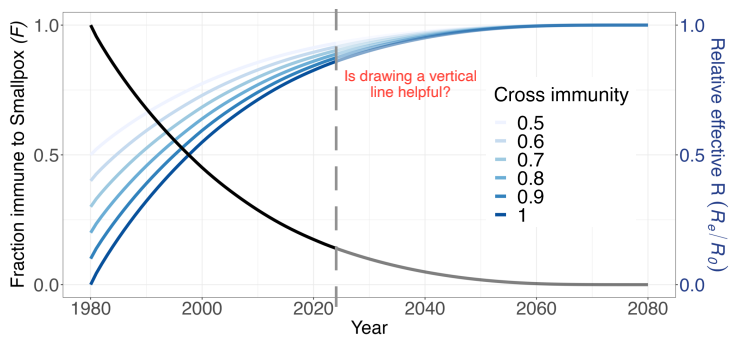

Figure 2: Waning smallpox immunity and the effective reproductive number scaled to its value in a naive population (i.e. R_e/R_0) for mpox. The left y-axis and black line indicates how the fraction of individuals in the DRC with immunity to smallpox wanes after vaccination was stopped in 1975. The right y-axis and blue lines indicate how the effective reproductive number of mpox transmission in the human population increases over time. Forward projections shown in lighter colors. We assume the fraction of the population immune to smallpox to equal the fraction of individuals born prior to 1980 (the date around which smallpox vaccination was stopped).

slightly less than 1 [8]. However more recent studies estimating R_0 from eqn 2 suggest that it is in the range of 1, and likely slightly greater than 1. These studies are based on estimation of R_e of mpox to be around 0.3 from the length of stuttering chains of transmission of mpox outbreaks in the human population between 1980 and 1984 [6], and estimates of $c \sim 0.85$ [13, 22] and $F \sim 0.9$ at this time.

Relationship between R_e and evolutionary emergence

The change in population immunity to mpox affects its emergence by increasing the number of spillovers and also the the number of secondary cases arising from each spillover. The latter increases the length of the chains of transmission and the probability of adaptive evolution that could lead to the emergence of a highly transmissible mpox variant capable of widespread circulation in the human population. We consider these separately below.

The rate of spillovers

The rate of spillovers per unit time, $S(t)$ equals the rate of exposures of individuals to the zoonoses (S_0) times the fraction of individuals who are susceptible to the infection. The fraction of susceptible individuals equals $(1 - F)$, where F equals the fraction immune. Thus as immunity wanes the number of spillovers gradually increases over time as

$$S(t) = S_0(1 - F(t)) \tag{3}$$

Is there no 'c' here? or is it assumed to be 1?

Probability of evolutionary emergence per introduction (P_E)

We define the probability of evolutionary emergence, P_E as the probability a single spillover results in the emergence of a highly transmissible variant of the pathogen. We use a branching process model of disease transmission (adapted from [3?]) to consider how P_E depens on R_e the reproductive number of the introduced virus. In this approach the introduction of an infection from the zoonotic reservoir is followed by stochastic chains of transmission with variable numbers of subsequent infections. If $R_e < 1$, in the absence of pathogen adaptation, these chains of transmission will stutter to extinction. Adaptive evolutionary changes (termed mutations) in the virus can however result in an increase in its reproductive number, R. The probability of emergence of a zoonotic infection from $R_e < 1$ (subcritical) to $R > 1$ (supercritical) depends on the reproductive number of the initial zoonotic infection (R_e), the beneficial mutation rate per infection (μ) (which we subsequently call simply the mutation rate) and the fitness advantage conferred by a mutation (σ). We incorporate two additional features to the earlier model. First, as the effective reproductive number R_e of mpox increases fewer adaptive mutations are required for the reproductive number of the evolved variant to exceed one. Second, we allow adaptive evolution to continue after emergence. We let this occur till the reproductive number of mpox reaches a value comparable to that of smallpox (i.e. $R_0 = 3$). We refer the reader to the supplement for the branching process calculations.

simply is extra

In Fig 3 we plot how the probability of evolutionary emergence depends on three factors: the initial reproductive number R_e, the rate of generation of adaptive mutants μ, and the selective advantage σ conferred by an adaptive mutation. In all the plots the reproductive number of the spillover, R_e is on the x-axis and the P_E (the probability of emergence) on the y-axis. P_E increases with increasing R_e. If $R_e > 1$, P_E is relatively high and only stochastic extinction when the numbers of infecteds is low prevents emergence. Fig 3A shows that P_E can fall very rapidly as R_e decreases below 1, and the dependence of P_E on R_e is stronger when the benefits arising from each mutation (σ) is small. This is because as σ decreases the number of mutations (n) required for R to exceed one increases as approximately $(1 - R_e)/\sigma$. Fig 3B shows how P_E declines as the mutation rate μ falls. When $R_e < 1$ the dependence of P_E on μ is compounded by the number of mutations required in order for R to exceed one as $\approx \mu^n$.

Evaluating pandemic potential.

We define the pandemic potential as the probability of a successful emergence event per unit time. The pandemic potential is the product of the spillover rate and the probability of emergence per spillover, P_E. We have considered the P_E in the previous section, and now consider the dependence of the probability of a pandemic on the number of spillovers. The probability of a pandemic after n spillovers equals $1 - (1 - P_E)^n$ which is approximately nP_E if $nP_E << 1$. This

Not clear how n is related to s?

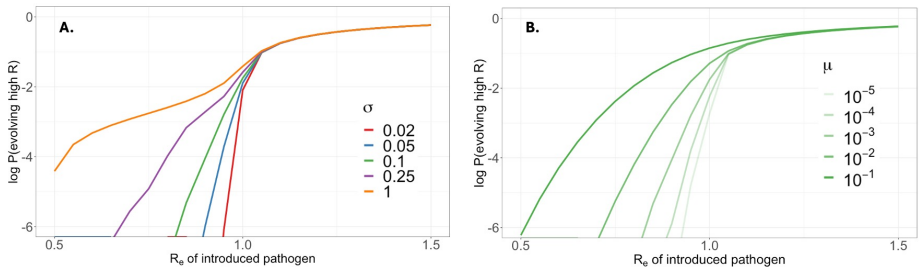

Figure 3: Dependence of the probability of evolutionary emergence per introduction (P_E) for a pathogen on the reproductive number R of the zoononoses (on the x-axis), and the selective advantage σ of a mutation (Panel A) and the mutation rate μ (Panel B). The probability of evolutionary emergence increases very steeply (the y-axis is on a log-scale) with R, and falls rapidly as σ and μ decrease. The calculations for the probability of evolutionary emergence (defined as reaching an $R_0 = 3$) are described in the SI.

is intuitive – when P_E is small the probability of a pandemic scales linearly with the number of spillover events and saturates as the the probability of a pandemic approaches one.

In Fig 4 shows the relative contribution of the extent of transmission, R_e, and the rate of spillovers s on the pandemic potential. We focus on R_e and s since these parameters change (increase) as population level immunity to smallpox wanes, and the mutation rate μ and the selective advantage σ per adaptive mutation do not. We see that the pandemic potential depends much more strongly on changes in R_e than in s. This is particularly evident if we note that s is shown on a logarithmic scale. The black lines indicate probabilities of emergence of between 0.01 and 0.001 per year which would correspond to average times of emergence of 100 and 1000 years. We see that there is a narrow range of values of R_e for which the probability of emergence lies in this range. This range of times to emergence and R_e characterize pathogens which could potentially emerge in a timeframe of relevance. If R_e is higher than this range emergence is imminent, and likely already has occurred. If R_e is lower than this range than the pathogen is very unlikely to emerge in an individuals lifespan and thus not in a timeframe of relevance to most individuals and beyond the concern or even interest of policymakers or politicians. Consequently few pathogens might be expected to fall in this regime for which evolutionary emergence is a concern. One exception to this is mpox. This is because with waning immunity to smallpox the R_e for mpox is almost certain to go from a value much less than one (about 0.3 in the 1980's) to a value around or slightly greater than one as the fraction of individuals with immunity to smallpox falls to close to zero in the next decade or two.

This assumes no epistasis? If yes, how valid is that assumption?

No grey lines in Fig 4! I see grey areas.

The grey lines on Fig4 indicate changes in R_e and S as immunity to smallpox wanes. We choose to begin at the $R_e = 0.3$ from the estimate in the early 1980's and show how the the pandemic potential changes with time thereafter for a range of final values of R_0 of mpox ranging from 0.95 to 1.2. We see that the change in the pandemic potential due to waning immunity is largely driven by

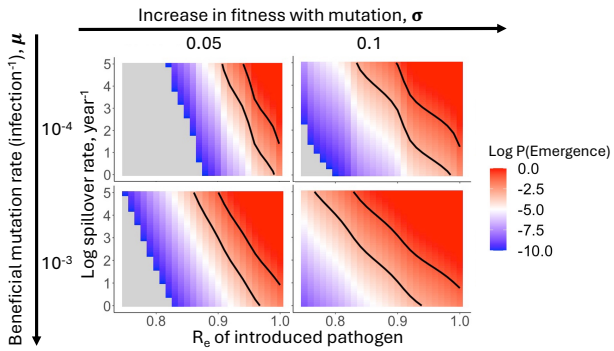

Figure 4: **Pandemic potential**. We explore the pandemic potential for a pathogen – the probability that in one year introductions lead to the evolutionary emergence of a variant that can spread rapidly in the human population. The top row corresponds to XX spillovers per year and the bottom row to 100 fold increase in this rate. We see that increasing the rate of spillovers (going from the top to bottom panel) does not have nearly as large an effect on the pandemic potential as changes in the parameters μ and σ that describe the evolutionary potential of the virus or the initial transmissibility of the virus R. The red lines correspond to a pandemic potential of 1 per 10 and 1 per 10^3 years. I don't see any red lines!

I thought the rows represent different mutation rates.

increases in R_e rather than the change in the spillover rate.

Implications for intervention

We have identified four parameters that play an important role in governing the pandemic potential of an emerging virus. These are the mutation rate μ, the adaptive benefit caused by a mutation σ, the transmissibility of the zoonosis R and the rate of spillover events s. Two of these parameters, μ and σ are properties of the pathogen-host system and are not under our control. Consequently we focus on how different interventions can reduce the pandemic potential of a pathogen by decreasing transmissibility (R) and the rate of spillovers (s).

Immunization can reduce both the rate of spillovers and the extent of subsequent transmission of the pathogen in the human population, and as described earlier the main contribution to reducing pandemic potential will be by the reduction in transmission afforded by the vaccine. Vaccines can reduce transmission by susceptibility of individuals to infection or their infectivity if they do get infected. Reducing pathology, while beneficial to the individual has no direct effect on pandemic potential, but can have indirect effects which will be discussed late.

Having an accessible and efficient public-health and clinical healthcare system can reducing pandemic potential by a reduction in transmissibility R. This could be accomplished by rapid testing, appropriate treatment reducing the duration of infection and quarantine measures as applicable.

A final way of reducing pandemic potential and emergence is by reducing the ecological overlap between the zoonotic reservoir and human hosts [17]. We

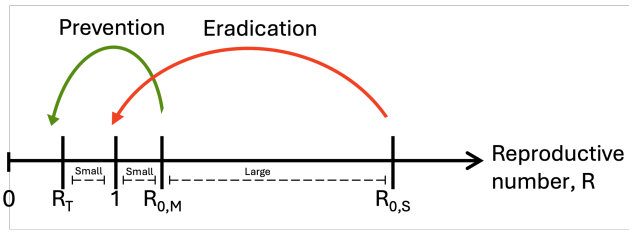

Figure 5: **Delayed Intervention.** The magnitude of response required to control disease after evolutionary emergence(1-1/R_0) to high transmissibility may be significantly greater than the effort required to prevent emergence (1-R_T/R_0, where R_T is some R at which the probability of evolutionary emergence is low) when $R_{0,M}$ is near (slightly less than or slightly greater than) one.

suggest this is likely to have a minimal effect on the emergence of mpox. While our results suggest a proportional reduction in s has a much smaller effect than similar reduction in R, this does not take into account the costs to achieve similar fold reductions in these parameters, the potential for the reductions to cover multiple pathogens, or have other beneficial effects.

Costs of delaying intervention

It is widely held that prevention is better than a cure and this is true for the case of mpox, as we illustrate in Fig 5. In its simplest form prevention might occur by ensuring R_e (or more accurately combination of R_e and S) is kept below a critical level that corresponds to an acceptably low probability for evolutionary emergence as depicted in Fig 5. We see that for a wide range of parameters this requires keeping R_e to a level slightly below one (the level depends on σ, and μ). This might entail, order of magnitude a 20% reduction in transmission (if for example it requires reducing R_0 of the zoonotic infection from 1 to 0.8). In contrast if mpox for example has had time to adapt to a level commensurate with smallpox which has an R_0 about 3 then pathogen elimination requires preventing $1 - 1/R_0 = 2/3$ or 66% of all transmissions. This is shown in Fig 5 which illustrates that a much lower level of intervention such as vaccination is needed to prevents a pandemic rather than control it after it has occurred.

2 Discussion

Cross-immunity arising from natural infection or vaccine-induced immunity to smallpox largely prevented zoonotic spillovers and human-human transmission of mpox. The end of smallpox circulation and vaccination in 1980 has led to concomitant waning of cross-immunity to mpox and an increase in both the number of zoonotic cases and the human-human transmission of mpox. We have used mathematical models to explore the interplay between waning immunity and evolutionary adaptation and applied this theory to the potential emergence of an evolved and highly transmissible mpox virus, and note its more general

applicability to the emergence of zoonotic pathogens following the eradication of related endemic disease.

Our analysis required extending a previous model of the evolution of a pathogen during stuttering chains of transmission after spillover in three ways. First we incorporated changes (an increase) in the effective reproductive number (R_e) for a single pathogen (mpox) as cross-immunity waned. Second, we integrated the increase in R_e with a decreases in the number of mutations needed for the virus transmission to become supercritical. Finally we allowed evolution continue even after the reproductive number after the virus in the human population was greater than one.

Our analysis showed that as R approaches one there is a very rapid rise in the probability of evolutionary emergence, and this steep relationship is exacerbated by the decline in the number of mutations required as R approaches one. The rapid change in the probability of emergence with reproductive number suggests that there is only a small range of R for which emergence is likely to be a concern – if the probability of emergence is too small then it is not likely to occur in say the next 100+ years, and if it is very high then the pathogen will have emerged or there will be little to prevent it emerging unless drastic measures are taken to reduce its reproductive number. The waning of immunity to mpox will result in its reproductive number increasing from well below one in the 1980's (estimated at around 0.3) to close to or likely above one in the next decade when the population has almost no immunity to smallpox or mpox. Indeed, recent outbreaks of mpox have been large and potentially sustained [21]. Thus mpox may well be passing through the regime where emergence is very likely now or in the near future.

Interventions to prevent the evolutionary emergence of mpox can do so by reducing the frequency of spillover infections (s) or the extent of transmission in the human population. Our results show that the latter (decreasing R) is much more effective effective than the former (decreasing s). Ecological interventions that reduce contact between the zoonotic reservoir and humans can certainly reduce the number of zoonotic infections but unless the number of spillover events is very low our results suggest that this is unlikely to have a substantial effect on the probability of evolutionary emergence. Other interventions such as improvements in surveillance and universal access healthcare can reduce the transmission of emergent pathogens such as mpox in humans and have the advantage of doing so not just for mpox, but also for other zoonoses and would benefiting many aspects of the health of communities. Vaccination can decrease evolutionary emergence by reducing both the spillover rate and transmission of a disease, and our study suggests that its effect is largely from the extent of reduction in transmission. Vaccination can work in multiple ways – it can reduce the susceptibility of individuals to infection, and it can reduce the infectivity and the extent of pathology in those that do get infected. Preventing the evolutionary emergence of mpox by vaccination would benefit from the design of a vaccine that confers broad orthopox immunity potentially at the expense of immunity to smallpox and would also require a focus on reducing transmission (by a combination of reducing susceptibility and infectivity) rather

you mean pandemic potential.

than pathology. Indeed a vaccine that only reduced pathology might increase the probability of emergence due to transmission from asymptomatic infections. Indeed it may even be that permitting some symptoms can allow for the better detection of spillover infections and facilitate measures that reduce transmission [9].

Perhaps the most salient result is the observation that the level of vaccination or other intervention required to prevent emergence is modest compared to that needed for control and eradication after it has adapted to high transmissibility in the human population post-emergence. Due to the persistence of the pathogen in zoonotic reservoirs, however, this low-level vaccination needs to be ongoing and introduces an interesting difficulty with the counterfactual. To what extent can we know the true necessity of prevention efforts if the disease never emerges? We leave this question to the reader as a philosophical exercise.

2.1 Caveats, Limitations and Future Directions

We have a relatively poor understanding of the emergence of new pathogens into the human population, and there are a number of reasons for this including limitations in our understanding of the evolutionary and epidemiological processes involved and the paucity of examples.

We have a limited understanding of the mutational and adaptive landscapes in general, and the extent to which changes in the pathogen in the zoonotic reservoir result in random changes in its transmissibility in humans. In the case of mpox this might require understanding how these viruses can adapt to evade immunity [11**?**].

There have only been a handful of pathogens that have successfully emerged into the human population and of these only a few recently enough to allow collection and analysis of samples of the pathogen and data and analysis on its spread. Examples include HIV, new influenza subtypes H1N1 in 1918 and 2009, H2N2 in 1957 and H3N2 in 1968, SARS-CoV-2 in 2019 and Zika together with a few incipient emergence events (give examples SARS-CoV-1, MERS, Ebola that have been controlled. Even in these cases there is limited data on events at the time of emergence (when the number of cases is small). In the case of SARS-CoV-2 there considerable controversy on the role of natural vs laboratory manipulations played on its evolutionary emergence.

In addition to the limitations in our understanding of potentially emerging pathogens there are complexities associated with their transmission in the human population. These include the heterogeneity which has been the subject of considerable interest, and is a potential direction for extensions of the current model. In the case of HIV for example, the persistent nature of the infection resulted in highly connected nodes of sexual networks contributing substantially to its spread, while in the case of mpox which is an acute infection individuals in highly connected nodes would become immune potentially limiting further spread. Another problem is that the spread of an emerging pathogen can result in changes in the contact patterns of humans and modify its spread, and the extent of these changes can vary dramatically between different locations as was

seen following the initial emergence of SARS-CoV-2 where some countries could prevent successful introductions (e.g. New Zeeland), a few could introduce sufficiently severe lock-downs that resulted in local eradication (e.g. China) and most could at best flatten or reduce the number of cases prior to development of a vaccine.

While the formulation of a critical threshold for vaccination $f > 1 - 1/R_0$ provided the rationale for the eradication of smallpox, actual eradication required a far more nuanced and complex strategy than included surveillance and case detection followed by ring vaccination. Similarly the considerations outlined in this paper might inform us of how early and ongoing intervention might be needed following the eradication of pathogens such as smallpox.

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Preventing disease emergence following eradication: Application to mpox

1 Multi-type branching process formulation

Our model for the adaptive evolution of a zoonotic pathogen after spillover into humans is based on a previously developed multi-type branching process model of emerging infectious diseases (**??**). The multi-type branching process formulation allows for an easy derivation for the probability of emergence.

Consider the case where a pathogen must acummulate a number of mutations after spillover into the human population to attain $R_0 > 1$. Given m variants from the wildtype to the fully-evolved mutant with $m-1$ mutations required to attain $R_0 > 1$, the m-type branching process can be described by m probability generating functions (PGFs):

$$f_i(s_1, s_2, ..., s_m) = \sum_{j_1, j_2, ..., j_m = 0}^{\infty} p_i(j_1, j_2, ..., j_m) s_1^{j_1} s_2^{j_2} ... s_m^{j_m}, \ i \in \{1, 2, ..., m\}$$

where $p_i(j_1, j_2, ..., j_n, ..., j_m)$ is the probability that an infection of type i gives rise to j_n secondary infections of type n.

The probability of extinction q_i of a transmission chain initiated by a single infection with a variant of type i is given by the least fixed-point solution to: $f_i(q_1, q_2, ...q_m) = q_i, \ i \in \{1, 2, ..., m\}$.[1]

We define a model in which there exists m different variants of the pathogen, with one wildtype and $m-1$ mutants. We denote the basic reproductive number of the wildtype pathogen as $R_0^{(1)}$, the basic reproductive numbers of the $m-2$ intermediate mutants as $R_0^{(2)}, ..., R_0^{(m-1)}$, and the basic reproductive number of the fully-evolved mutant as $R_0^{(m)}$.

We assume the following about mutation and the spread of infections:

[1] Note that this is a special case of the generalized expression for the probability that a process initiated with $r_1, r_2, ..., r_n$ chains of infections of type $1, 2, ..., m$ goes extinct is given by the product: $q_1^{r_1} q_2^{r_2} ... q_m^{r_m}$. Ref: Notes from Linda Allen...

- the mutation rate, μ, of all the variants is the same
- only single variants occur, i.e the itth variant can only mutate into an $i+1$ mutant
- the total number of secondary infections arising from an individual infeected with variant i is Poisson-distributed with mean $R_0^{(i)}$
- the number of secondary infections of type i arising from an individual infected with variant i is Poisson-distributed with mean $(1-\mu)R_0^{(i)}$
- the number of secondary infections of type $i+1$ arising from an individual infected with variant i is Poisson-distributed with mean $\mu R_0^{(i)}$

These assumptions define the PGFs $f_1, f_2, ..., f_m$, given by:

$$f_i(s_1, s_2, ..., s_m) = \exp[(1-\mu)R_0^i(s_i-1)] \exp[\mu R_0^i(s_{i+1}-1)],\ i \in \{1, 2, ..., m-1\}$$
$$f_m(s_1, s_2, ..., s_m) = \exp[R_0^m(s_m-1)]$$

The probability of emergence can be calculated numerically from the PGFs according to the fixed point equation above.

Our model makes 2 important departures from the previous model:

1. The fitness of intermediate variants increases by some proportional amount, σ;
2. Evolution continues to increase the reproductive number after the first $R > 1$ variant.

We alter the framework presented in **??** according to the following.

Consider the case of adaptive evolution during transmission. After the spillover infection, mutations are acquired during subsequent transmission events, each increasing the R of infections by some proportion, σ, until a variant emerges with an $R_0^{(m)}$ exceeding some $R_0^{(m')}$.

The number of mutations $m-1$ in the lineage from the wild-type (1) to the fully-adapted type (m) thus depends on the reproductive number of the wildtype $R_0^{(1)}$, the minimum reproductive number of the fully-adapted type $R_0^{(m')}$, and the proportional increase in $R_0^{(i)}$ to $R_0^{(i+1)}$ with mutation and is given by:

$$m-1 = \frac{log(R_0^{(1)} - log(R_0^{(m')})}{log(1+\sigma)}$$

Note that $R_0^{(m)}$ is the first R_0 that is greater than the minimum $R_0^{(m')}$.

To calculate the probability that the lineage does not go extinct before exceeding $R_0^{(m')}$, we replace the final $R_0^{(m)}$ with some R arbitrarily high such that the contribution of extinction of the final type is negligibly small (once one of that type arises the process will never go extinct.)

(stochastic simulations that match this if we need to address that)

Fin.